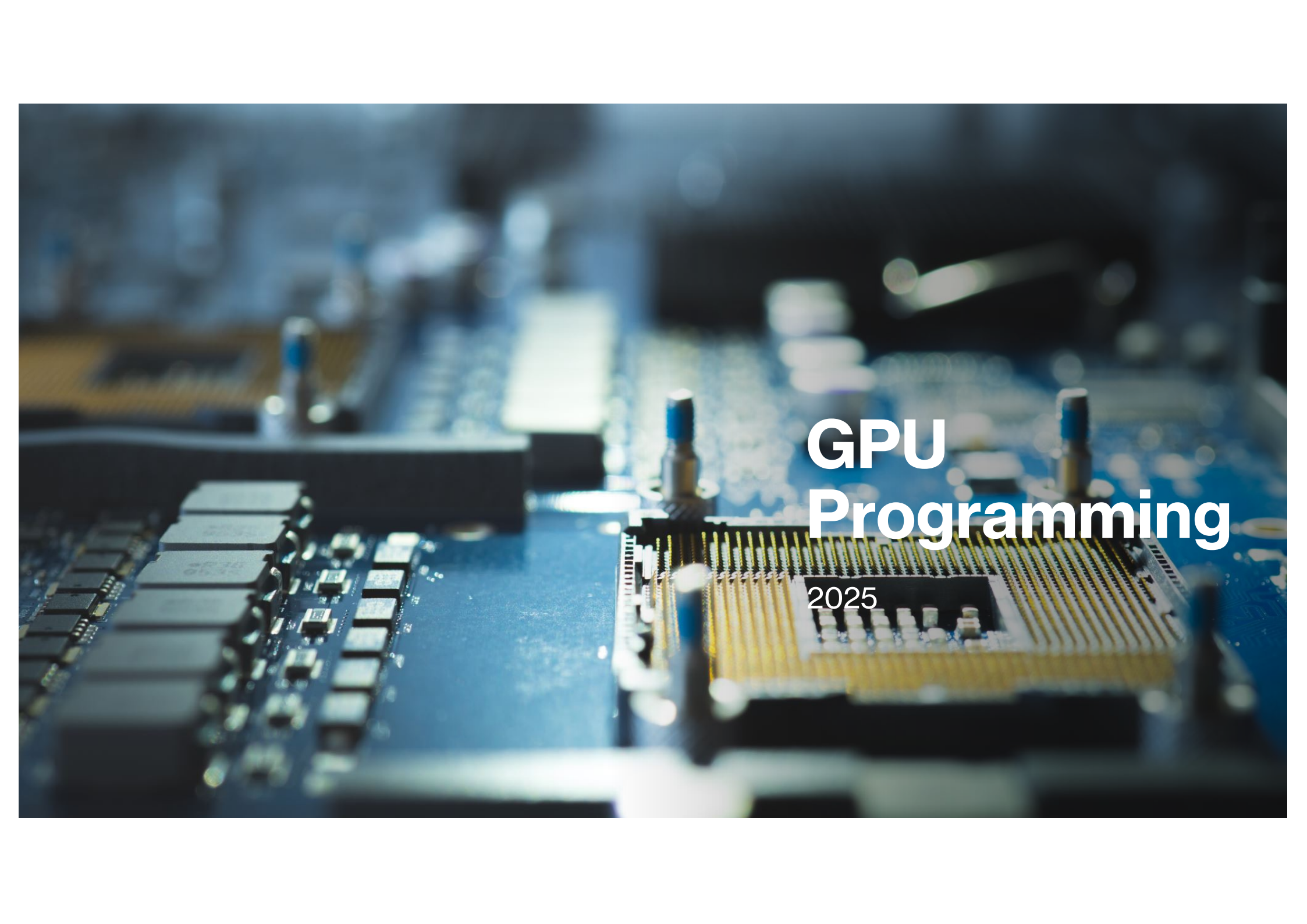


A close-up, shallow depth-of-field photograph of a GPU (Graphics Processing Unit) mounted on a blue printed circuit board (PCB). The GPU is a square chip with numerous gold-plated pins along its edges. To the left of the GPU, there are several rows of memory modules (likely GDDR6) also mounted on the PCB. The background is blurred, showing other components and the overall structure of the board. The lighting is soft, highlighting the metallic surfaces of the components.

GPU Programming

2025

NOTE

- Course Instructor **Jyothi Vedurada** is on an academic leave for some important academic work this whole week.
- TAs will present the same introduction slides (as today's) in Wednesday's class (30th July).
- **No class on Thursday** (31st July)

Logistics

- Google classroom for assignment submissions, announcements and discussions.
 - Make sure from your side that you are added to the classroom: [tdjrxec7 \(Google classroom code\)](#)
- Evaluation:
 - End-Sem: 25%
 - Assignments: 30% (3 assignments)
 - Project: 20% (1.5 – 2 months) (in groups but marks vary for each member based on contributions)
 - Mid-Sem: 10%
 - Surprise Quizzes: (3) 10%
 - Class participation: 5%
- C slot (**Monday 11 am. Wednesday 10 AM. Thursday 9 AM.**)

Logistics (Contd..)

- Tas:
 - PhDs: Soumik, Karthik and Pirapuraj
 - Mtechs: Chinmay and Siddhartha
- Contact Hours:
 - Instructor: **Wednesday (2:30pm to 5:30pm)**
 - TAs: **Friday (5:00pm to 7:00pm)**
- GPUs:
 - Assignments and project requires running code on GPU
 - You need to have your own GPU in your laptop or workstation or your lab machine
 - Otherwise use Cloud resources: Google Cloud, AWS student credits

Logistics (Contd..)

- Required Skill set
 - Very strong coding skills in C and C++
 - Compilers background: The course may cover compiler static analyses for CPU-GPU systems which requires implementing analysis in LLVM framework (millions of lines of code)
- Course is heavy in implementation of assignments+projects as well as theory exams
- No audit allowed. Even if someone audits on instructor's permission, they have to submit assignments and write exams.

Course Contents

1. **Introduction to GPU Programming:** Evolution of GPUs, heterogeneous computing, overview of CUDA, OpenCL, and other parallel programming interfaces
2. **GPU Programming Fundamentals:** CUDA program structure, kernel execution, threading model (grids, blocks, warps), Memory hierarchy (global, shared, constant), synchronization, Control divergence
3. **Performance Optimization:** Thread coarsening, memory coalescing, occupancy, and latency hiding, Profiling
4. **Parallel Algorithms & Patterns:** Reduction, scan (prefix sum), stencil computation, and convolution, Parallel sorting (radix, merge) and graph algorithms (BFS, edge/vertex-centric), sparse matrix-multiplication
5. **Advanced GPU Computing:** Multi-GPU programming, CUDA streams, dynamic parallelism, and unified virtual memory
6. **Applications & Case Studies:** Image/signal processing, large-scale graph/search algorithms, simulation, and deep learning
7. **Static Analyses on GPU:** Thread divergence analysis, analyses to compute memory coalescing

Other points

- Evaluation is strict. No liberal marks.
- Class participation which includes attending classes and participating in the classes (asking questions and answering questions) is taken seriously.
- Plagiarism or any kind of malpractice: Direct 'F' grade and will be sent to disciplinary committee.
 - Moss (static analysis tool from Stanford) and LLM models would be run on all the submitted assignments and projects to perform code similarity checks.
- Any use of AI models should be cited/mentioned appropriately when submitting the assignments.
 - Viva is conducted in doubtful cases, and an 'F' grade is given if found direct copy from AI models' output.

Homework for this week

- Figure out a GPU for doing your assignments and project
- Install CUDA and keep the setup ready